



Department of Information Technology
Academic Year 2025– 2026(Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic: Unit 2/ Proving languages to be not regular**

Course Outcome: The student will be able to solve regular expression for any regular language and find equivalence of regular expression and finite automata.

- **Topic Learning Outcome: TLO6.**

- **Activity Chosen: Cross Word Puzzle**

- **Justification:**

- The concept of Proving languages to be not regular is one of the important topics and the question from this topic asked in most of the University questions. To understand the level of students this activity is planned.

- **Time Allotted for the Activity: 10 Minutes**

- **Details of the Implementation:**

- In the previous day class was taken about proving languages to be not regular.
- Ask the students to prepare well for the topic.
- The puzzle was displayed on the screen.
- The Question and Answer for the topic as shown in Figure 1.
- The sample presentation of the student Mr.R.Suresh as shown in Figure 2 and 3.

- **CO – PO / PSO mapping:**

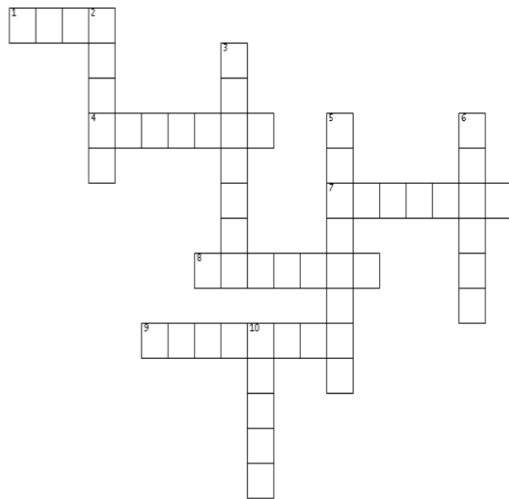
CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	2	3	1	2	1	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	2	3	1	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals Proving the language is not regular for finding the expression is correct or not	Identify the need of Proving the language is not regular for developing applications	Able to design solutions for implementing Proving the language is not regular.	Able to analyze the relationship between the language is regular or not	Able to prove the language is regular	Able to find the solution individually	Based on the given problem, the answer can communicate to a group of peoples	Ability to recognize the need for Proving the language is not regular in real world applications	Exhibit knowledge in Proving the language is not regular to provide solution for complex computational problems	Exhibit knowledge in Proving the language is not regular in the field Data Analytics to provide IT solutions.	Exhibit knowledge in Proving the language is not regular techniques in software projects to deliver a quality product

• **Images / Screenshot of the practice:**



ACROSS

1. Operation that represents repetition of a symbol any number of times
4. The empty string in formal language theory
7. Lemma used to prove that a language is not regular
8. Languages that can be expressed using regular expressions
9. A set of strings over an alphabet

DOWN

2. A notation used to describe patterns in strings
3. Property showing regular languages remain regular after operations
5. The set of symbols used to form strings in a language
6. Operation used to join two strings together
10. Operation that represents a choice between two expressions

Figure1: Questions and answer of Cross word puzzle

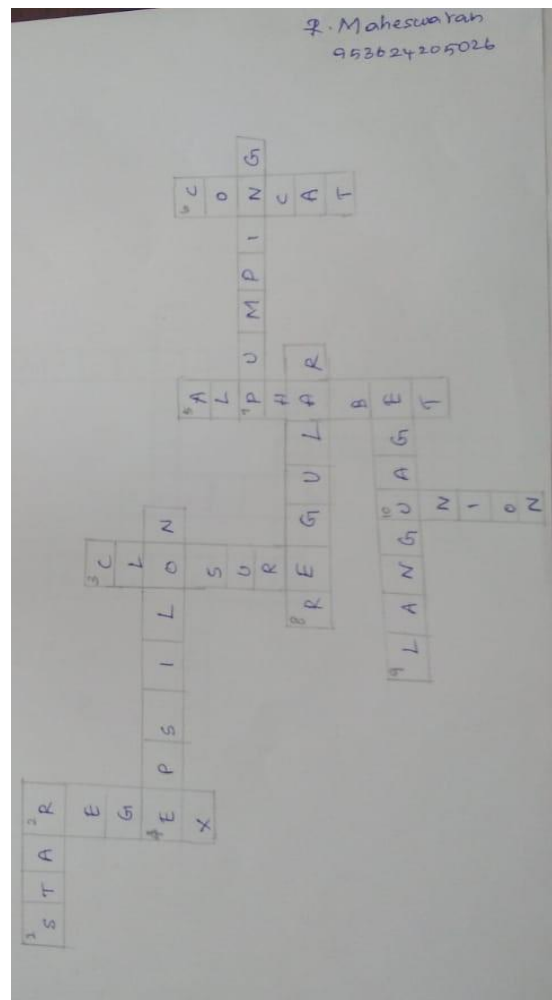
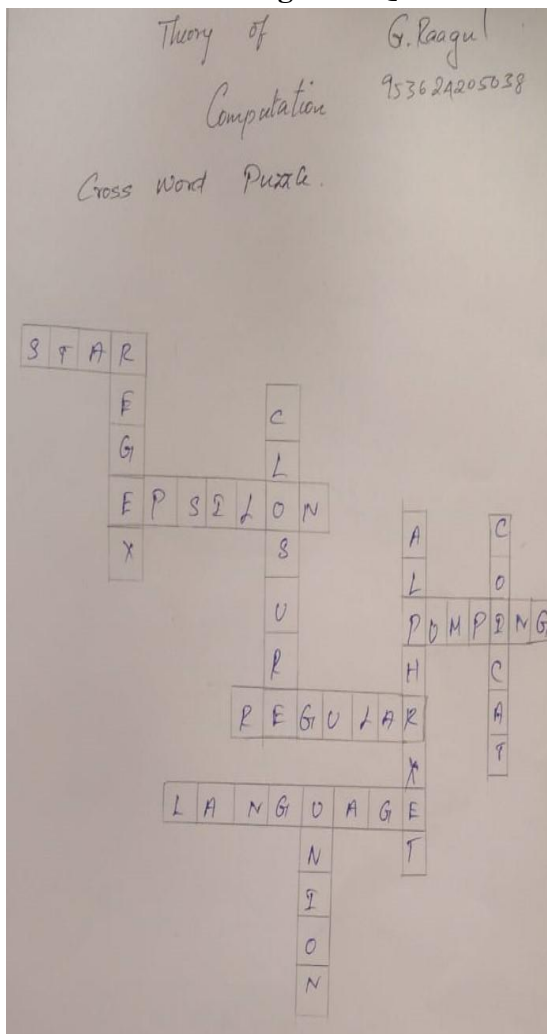


Figure 2: Answers given by the students Ragul G and Maheswaran R

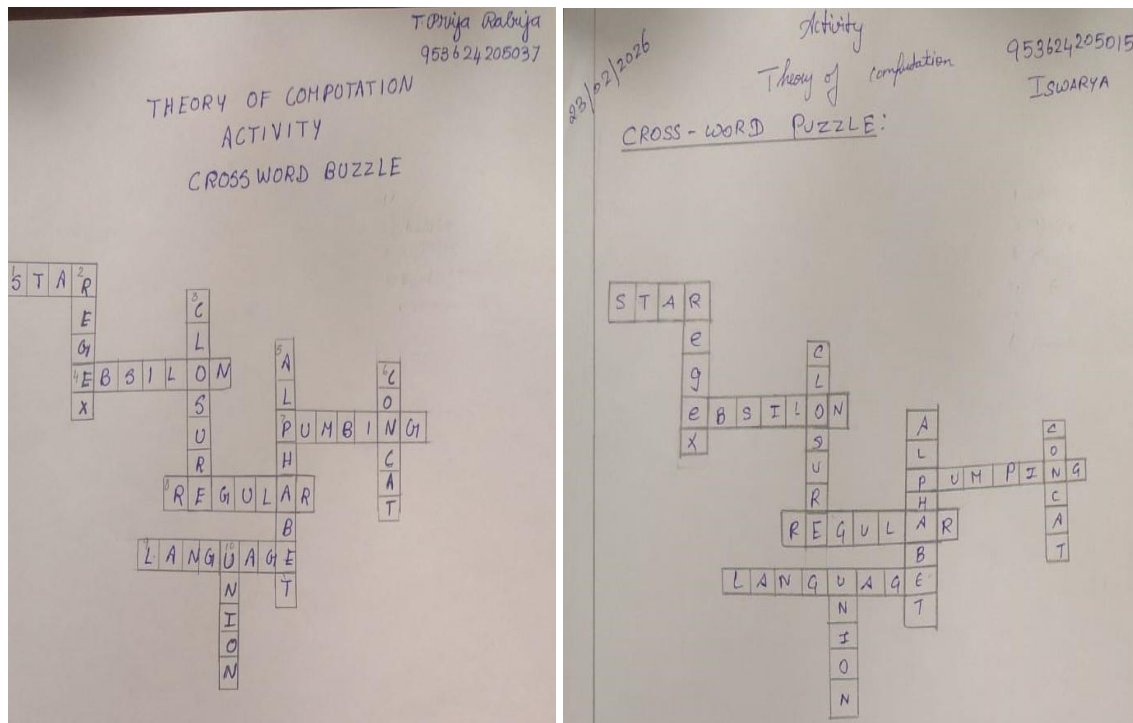


Figure 3: Answers given by students Priya Rabiya. T and Iswarya

- **Reflective Critique:**

- **Feedback of practice from students and other stakeholders:**

- Students actively participated in the crossword puzzle activity with enthusiasm.
 - The activity provided an opportunity for students to test and reinforce their understanding of concepts related to proving that a language is not regular (e.g., Pumping Lemma, closure properties).
 - Students expressed that the activity made abstract theoretical concepts more approachable and engaging.
 - Faculty observed improved interaction among students, promoting collaborative learning and peer discussion.
 - Overall, the activity created a positive and interactive learning environment.

- **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students developed a clearer conceptual understanding of techniques used to prove that a language is not regular.
 - The activity encouraged logical thinking and analytical reasoning skills.
 - Peer discussions reduced students' hesitation and fear in approaching complex theoretical topics
 - Students became more confident in applying theoretical concepts to problem-solving scenarios

- **Challenges faced in implementation:**

- Some students initially found it difficult to relate crossword clues to formal theoretical concepts.
 - Time constraints limited deeper discussion for certain complex clues.

- A few students required additional guidance to connect the activity with formal proof techniques like the Pumping Lemma.
- **References:**
 - https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/GM_CS3491_U1_CWP.pdf
 - https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/AAL_CS3551_Crossword_Puzzle.pdf
 - https://www.ritrjpm.ac.in/images/computer-science/2023-2024/IP/CS3391_GSP_23_24_CWP.pdf

Signature of Faculty Member

HOD